LCM Least Common Multiple no "greatest common multiple".
Prime factorisation of 2 to 10:
$2=2^1$, $3=3^1$, $4=2^2$, $5=5^1$, $6=2^1x3^1$, $7=7^1$,
$8=2^3$, $9=3^2$, $10=2^1x5^1$
LCM(2 to 10)$=2^3x3^2x5^1x7^1 = 2520$ (select highest powers)
Primes (Product of Primes = Prime Factorisation)
Primes 1 to 100: 2,3,5,7,11,13,17,19,23,29
,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97
If you see (mod n): often multiples/LCM — looking for x that is a multiple of n.
If you see f(n)=0: factors/roots — finding N to make equation zero (e.g., Rational Root Theorem to find possible root(s)).
Log and ln $\boldsymbol{log_x a \times \div b = log_x a \pm log_x b \mid b^{log_b(x)} = x}$
$\ln x^m = \ln x \qquad \log_x x^m = \log_x n \qquad (\ln x = \log_e x \,/\, e^x)$
$m \ln x = \ln x \qquad m \log_x x = \log_x n \qquad (\ln e = 1 \mid \ln 1 = 0)$
Class 1 | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.1 |
-- $\boldsymbol{7056_n - 4577_n = 2257_n}$
Convert to Base 10: $(7n^3 + 0n^2 + 5n^1 + 6n^0)_{10}$
$-(4n^3 + 5n^2 + 7n^1 + 7n^0)_{10} = (2n^3 + 2n^2 + 5n^1 + 7n^0)_{10}$
---Skip---$\Rightarrow n^3 - 7n^2 - 7n - 8 = 0$
$\Rightarrow n = 8$ (Use Calculator Menu 5 (Equation-Polynomial))
Therefore $n = 8$, as all other roots are Imaginary (i).■
Using Rational Root Theorem Factors of $-\frac{8}{1} = \pm1/\pm2/\pm4/\pm8$
Therefore $n = 8$, as n must be Positive $\mathbb{Z} > 7$.■
Rational Root Theorem
$\frac{Factors\ of\ Constant\ Term}{Factors\ of\ Leading\ Coefficent} = \frac{c}{L}\ (Lx^3 + x^2 + x + C)$
Factor of $\frac{c}{L}$ = All possible $\pm$ Roots (Answers)
Class 2
$\lim_{n\to\infty} \frac{1}{n} = 0$
Sum of the First nth terms of Arithmetic Series
$S_n = \frac{n}{2}(2a + (n-1)d)\ /\ S_n = \frac{n}{2}(a + c)$
$S_n = Total\ sun\ of\ first\ n\ terms\ /\ n = Number\ of\ terms$
$a = First\ Term\ /\ c = Last\ Term$
$d = Constant\ value\ added\ to\ each\ term\ to\ get\ next$
Nth term formula of Arithmetic Series
$T_n = a + (n-1)d$
Sum of the First nth Term of Geometric Series
$S_n = \frac{a(r^n-1)}{r-1}\ (r \neq 1)(If\ not\ it\ is\ Undefined)$
$S_n = Total\ sun\ of\ first\ n\ terms\ /\ n = Number\ of\ terms$
$a = First\ Term$
$r = Common\ value\ multipled\ to\ each\ term\ to\ get\ next$
Nth term of Geometric Series $T_n = ar^{n-1}$
Class 3 Special Sums
$\sum_{r=0}^{n} r = \sum_{r=1}^{n} r = \frac{n(n+1)}{2}$
$\sum_{r=0}^{n} r^2 = \sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$
$\sum_{r=0}^{n} r^3 = \sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$
$\sum_{r=1}^{4}(3r^3 + 4r + 1) = 3\sum_{r=1}^{4} r^3 + 4\sum_{r=1}^{4} r + \sum_{r=1}^{4} 1$
$= 3\left(\frac{4((4)+1)(2(4)+1)}{6}\right) + 4\left(\frac{4((4)+1)}{2}\right) + 4(1) = 134$■
P-series Test | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.1 |
$\sum_{n=k}^{\infty} \frac{1}{n^p}$ Converges if $p > 1$ / Diverges if $p \leq 1$
Geometric Series Test
$\sum_{n=1}^{\infty} r^n = \sum_{n=1}^{\infty}(\frac{2}{3})^n = \sum_{n=1}^{\infty}(\frac{2^n}{3^n})$
Comparison Test (Uses Geometric/P-series Test) $\boldsymbol{a_n, b_n \geq 0}$
Converges: $a_n \leq b_n, b_n$ converges, therefore a_n converges
Diverges: $a_n \geq b_n, b_n$ diverges, therefore a_n diverges
Limit Comparison Test (Uses Geometric/P-series Test)
$\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right)(where\ 0 < Limit < \infty)$
Converges: $a_n \leq b_n, b_n$ converges, therefore a_n converges
Diverges: $a_n \geq b_n, b_n$ diverges, therefore a_n diverges
If $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = 0$ (Convergence only if not Test Fails, try another)
If $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \infty\ (Does\ Not\ Exist)$
(Divergence only if not Test Fails, try another)
Ratio Test
$\lim_{n\to\infty}\left(\frac{a_{n+1}}{a_n}\right) = L < 1\ \sum a_n$ is convergent
$\lim_{n\to\infty}\left(\frac{a_{n+1}}{a_n}\right) = L > 1\ \sum a_n$ is divergent
$\lim_{n\to\infty}\left(\frac{a_{n+1}}{a_n}\right) = L = 1$ It is Inconclusive
Root Test (Get Rid of Power of n in $(a_n)^n$)
$\lim_{n\to\infty}(a_n)^{\frac{1}{n}} = L < 1\ \sum a_n$ is convergent
$\lim_{n\to\infty}(a_n)^{\frac{1}{n}} = L > 1\ \sum a_n$ is divergent
$\lim_{n\to\infty}(a_n)^{\frac{1}{n}} = L = 1$ It is Inconclusive
Divergence Test
$\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ is divergent
Alternating Series Test
$\sum_{n=1}^{\infty}(-1)^n(a_n) = \sum_{n=1}^{\infty}\frac{(-1)^n}{a_n}$
Or $\sum_{n=1}^{\infty}(-1)^{n+1}(a_n) = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{a_n}$
If $a_n \geq 0$ and $\lim_{n\to\infty} a_n = 0$ and $a_{n+1} \leq a_n$.
Then $\sum_{n=1}^{\infty}\frac{(-1)^n}{a_n}$ or $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{a_n}$ is Convergent.
Class 4 Prove by Induction that $2^n > n^2$ for all $n > 4, n \in \mathbb{Z}$
Let P(n) be the statement $2^n > n^2, n > 4, n \in \mathbb{Z}$
Base case: P(5) when $n = 5$
$P(5) = 2^5 > 5^2 = 32 > 25$
Hence P(5) is true as $32 > 25$ and $n > 4, n \in \mathbb{Z}$.
Using Inductive Hypothesis, assume P(k) for some $k > 4, k \in \mathbb{Z}$, that is: $2^k > k^2$
Now proving that P(k+1) is also true.
That is: $2^{k+1} > (k+1)^2$
LHS: 2^{k+1} and RHS: $(k+1)^2 = k^2 + 2k + 1$
$P(k) = 2^k > k^2$ (by Inductive Hypothesis)
Multiply P(k) by 2: $2(2^k) > 2k^2 = 2^{k+1} > 2k^2$ $(2^{k+1} = LHS)$
We know $k > 4$, so we multiply it by k: $k^2 > 4k$
$\Rightarrow 2^{k+1} > 2k^2 = 2^{k+1} > k^2 + k^2$
We then add k^2 to $k^2 > 4k$: $k^2 + k^2 > k^2 + 4k$
$= k^2 + k^2 > k^2 + 2k + 2k$
We know $k > 4$, so we multiply it by 2 instead: $2k > 8$
And we know $8 > 1$
Hence $2^{k+1} > k^2 + k^2 > k^2 + 2k + 2k$
$> k^2 + 2k + 8 > k^2 + 2k + 1$ $(k^2 + 2k + 1 = RHS)$
Therefore $2^{k+1} > k^2 + 2k + 1 = LHS > RHS$
Hence P(k+1) is true. Since P(5) is true, if P(k) is true
Then P(k+1) is also true. Therefore, by Mathematical Induction, P(n) is true for all $n > 4, n \in \mathbb{Z}$
Class 5 Modulo
$a \equiv b\ (mod\ m)\ /\ b \equiv a\ (mod\ m)\ (a \neq b)$
$a = b + km\ (k \in \mathbb{Z})$
$-8\ (mod\ 5) \equiv 2 = 2\ (mod\ 5) \equiv 2$
$8\ (mod\ 5) \equiv 3$
Multiplicative Inverse
$\frac{a}{b}\ (mod\ m) \equiv a\ (mod\ m) \times b^{-1}\ (mod\ m)$
$\not\equiv \frac{a\ (mod\ m)}{b\ (mod\ m)}$
$(b \times \mathbb{Z}_m)\ (mod\ m) \equiv 1 \Rightarrow \mathbb{Z}_m\ (mod\ m)$ will be the Multiplicative Inverse
Additive Inverse of $-15\ (mod\ 4) \equiv 1$ is
$15\ (mod\ 4) \equiv 3$ where $(-15 \times 3)\ (mod\ 4) \equiv 0$
Additive Inverse is $3\ (mod\ 4)$.■
Fermat's Little Theorem
Find $3^{123}\ mod\ 5$, choose $a = 3, p = 5$
$a^{p-1} \equiv 1\ mod\ p$
By Fermat's Little Theorem: $3^{5-1} \equiv 1\ mod\ 5$
$3^4 \equiv 1\ mod\ 5 \Rightarrow Take\ to\ powers\ 123\ mod\ 4 \equiv 3$
$3^{123}\ mod\ 5 \equiv 3^3\ mod\ 5 \equiv 27\ mod\ 5 \equiv 2$■
Class 6 Kinematics
Scalar (Magnitude): Distance, Speed
Vector (Magnitude and Change in Direction): Velocity
Speed (m/s): $\frac{Distance}{Time}$
Velocity v(t) (m/s): $\frac{Displacement}{Time}$
Acceleration a(t) (m/s^2): $\frac{Change\ in\ Velocity}{Time}$
Differentiation: $n(t) \Rightarrow \frac{d(n)}{d(t)} = v(t) \Rightarrow \frac{d^2(n)}{d(t)^2} = a(t)$
V (m/s) = Final velocity / U (m/s) = Initial velocity
A (m/s^2) = Acceleration / T (s) = Time
S (m) = Displacement
$v = u + at$
$s = ut + \frac{1}{2}at^2$
$s = \frac{1}{2} \times (u + v)t$
$v^2 = u^2 + 2as$
$s = vt - \frac{1}{2}at^2$
Gravity $= 9.81m/s^2$ / $-9.81m/s^2$ (Usually Negative)
Class 7 $\boldsymbol{\frac{1}{cos\,x} = sec\,x \mid \frac{cos\,x}{sin\,x} = cot\,x \mid \frac{sin\,x}{cos\,x} = tan\,x}$
Cosine Rule: $\boldsymbol{c^2 = a^2 + b^2 - 2ab\,cos(C)}$
Sine Rule: $\boldsymbol{\frac{a}{sinA} = \frac{b}{sinb} = \frac{c}{sinc}}$
Area of Triangle (180°): $\boldsymbol{\frac{1}{2}ab\,sin\,(C)}$
Acute Angle: sin(Q)
Obtuse Angle: $\boldsymbol{sin(180° - Q)}$
Right Angle Triangle: H – Longest Side / O – Opposite of Q
Sin $= \frac{Opposite\ (O)}{Hypotenuse\ (H)}$ / Cos $= \frac{Adjacent}{Hypotenuse\ (H)}$ / $\sin^2 x + \cos^2 x = 1$
Sum To Product:
Sine Sum: $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$
Sine Difference: $\sin A - \sin B = 2\cos(\frac{A+B}{2})\sin(\frac{A-B}{2})$
Cosine Sum: $\cos A + \cos B = 2\cos(\frac{A+B}{2})\cos(\frac{A-B}{2})$
Cosine Difference: $\cos A - \cos B = -2\sin(\frac{A+B}{2})\sin(\frac{A-B}{2})$
Calculator Tricks: (x is the first answer in your Calculator)
$\cos x \Rightarrow 360° - x\ /\ \sin x \Rightarrow 180° - x\ (Third\ Angle{:}\ x + 360°)$ | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.1 |
$\tan x \Rightarrow 180° + x\ (Third\ Angle{:}\ (180° + x) + 180°)$
More Angles if it is within 360°
Quadratic Formula:
$\frac{-b\pm\sqrt{D}}{2a}$ $Discriminant\ (D){:}\ b^2 - 4ac$ (for $\mathbb{R}{:}\ D \geq 0$)
If $a > 0\ (Postive\ A)\ y \leq smaller\ root$ **or** $y \geq larger\ root$
If $-a < 0\ (Negative\ A)\ smaller\ root \leq y \leq larger\ root$
Consider the given $f(x) = \frac{x-3}{x^2+9x-22}$ Find Domain/Range
Since $f(x) = \frac{x-3}{x^2+9x-22}$ the denominator $x^2 + 9x - 22 \neq 0$
$(x + 11)(x - 2) \neq 0$
Therefore, the Domain $D_f = \{x \in \mathbb{R} \mid x \neq -11\ and\ 2\}$
Let $y = \frac{x-3}{x^2+9x-22}$ (Use this to find the Discriminant)
---Skip---$\Rightarrow yx^2 + x(9y - 1) + (3 - 22y) = 0$
$Since\ x \in \mathbb{R}{:}\ b^2 - 4ac \geq 0$
$\Rightarrow (9y - 1)^2 - 4(y)(3 - 22y) \geq 0$
$169y^2 - 30y + 1 \geq 0$ (sub into $\frac{-b\pm\sqrt{b^2-4ac}}{2a}$)
$\Rightarrow y = \frac{-(-30)\pm\sqrt{(-30)^2-4(169)(1)}}{2(169)}\ (a = 169 > 0\ (Positive\ A))$
Therefore, Range $R_f = \{y \in \mathbb{R} \mid y \leq \frac{15-2\sqrt{14}}{169}\ or\ y \geq \frac{15+2\sqrt{14}}{169}\}$
Horizontal line test:
Injective (one-to-one): Any horizontal line intersects the graph at most one time.
Surjective (onto): Every horizontal line intersects the graph at least once.
Bijective: Both injective and surjective it is Bijective hence it is invertible.
Class 8 Differentiation: $\boldsymbol{\frac{dy}{dx} \Rightarrow \frac{d^2y}{dx^2} \Rightarrow \frac{d^3y}{dx^3}}$
$y = f(x) \pm g(x)\ /\ \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$ (Differentiate Both)
$y = kf(x) = k \times f(x)\ /\ k \times \frac{d}{dx}f(x)$ (k is untouched $(k \neq 0)$)
$\sin x \Rightarrow \cos x \Rightarrow -\sin x \Rightarrow -\cos x \Rightarrow (Back\ to)\ \sin x$
$\cos x \Rightarrow -\sin x \Rightarrow -\cos x \Rightarrow \sin x \Rightarrow (Back\ to)\ \cos x$
$\tan x \Rightarrow \sec^2 x \Rightarrow 2\sec^2 x \tan x \Rightarrow 4\sec^2 x \tan^2 x + 2\sec^4 x$
$\ln x \Rightarrow \frac{1}{x} \Rightarrow -\frac{1}{x^2} \Rightarrow \frac{2}{x^3} \Rightarrow -\frac{6}{x^4} \Rightarrow \frac{24}{x^5}$
$log_{10}x \Rightarrow \frac{1}{x\ln 10} \Rightarrow -\frac{1}{x^2\ln 10} \Rightarrow \frac{2}{x^3\ln 10} \Rightarrow -\frac{6}{x^4\ln 10} \Rightarrow \frac{24}{x^5\ln 10}$
$e^{-2x} \Rightarrow -2e^{-2x} \Rightarrow 4e^{-2x} \Rightarrow -8e^{-2x} \Rightarrow 16e^{-2x}$
$(e^x \Rightarrow e^x\ Never\ changes)$
Product Rule:
$y = f(x) \cdot g(x)\ /\ f(x)\left(\frac{d}{dx}g(x)\right) + g(x)\left(\frac{d}{dx}f(x)\right)$
$y = x \cdot \ln x\ /\ \frac{dy}{dx} = x\left(\frac{1}{x}\right) + \ln x\,(1) = 1 + \ln x$
Quotient Rule:
$y = \frac{f(x)}{g(x)}\ /\ \frac{g(x)\left(\frac{d}{dx}f(x)\right) - f(x)\left(\frac{d}{dx}g(x)\right)}{(g(x))^2}$
The Hyper Tower:
$y = x^x \Rightarrow \ln y = \ln x^x \Rightarrow \ln y = x \ln x$
$\frac{d}{dx}(\ln y) = x\frac{d}{dx}(\ln x) + \ln x\frac{d}{dx}(x)$
$\frac{1}{y}\left(\frac{dy}{dx}\right) = x\left(\frac{1}{x}\right) + \ln x\,(1) = y(1 + \ln x) = x^x(1 + \ln x)$
Stationary/Critical Points of $y = 2x^3 - 6x^2 - 18x$
Set $\frac{dy}{dx} = f(x) = 6x^2 - 12x - 18 = 0$
$\Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$
When $x = -1, y = 2(-1)^3 - 6(-1)^2 - 18(-1) = 10$
When $x = 3, y = 2(3)^3 - 6(3)^2 - 18(3) = -54$
If $\frac{d^2y}{dx^2} < 0$: Local **Maximum** Point
If $\frac{d^2y}{dx^2} > 0$: Local Minimum Point
$\frac{d^2y}{dx^2} = 12x - 12 = 0$
When $x = -1, y = 12(-1) - 12 = -24 < 0$
When $x = 3, y = 12(3) - 12 = 24 > 0$
Therefore (-1, 10) is a Local Maximum Point and (3, -54) is a Local Minimum Point.■
Extreme Value Theorem: Find the Global Max and Min for the above $when - 1 \leq x \leq 3\ is\ given$
When $x = -1, y = 2(-1)^3 - 6(-1)^2 - 18(-1) = 10$
When $x = 3, y = 2(3)^3 - 6(3)^2 - 18(3) = -54$
Therefore, based on the Extreme Value Theorem the Global Minimum Point is (3, -54) and the Global Maximum Point is (-1, 10) which are stationary points of the graph.■
$\boldsymbol{\frac{dy}{dx}}$ ***Test*:** Test $x = -0.1, 0, 0.1$ in $\frac{dy}{dx} = f(x)$
If Positive is (+), Negative (-), _/ Local Min, /‾\ Local Max
Find and classify stationary points of $y = x^3$
Set $\frac{dy}{dx} = 3x^2 = 0 \Rightarrow x^2 = x(x) = 0$
$\frac{d^2y}{dx^2} = 6x \Rightarrow x = 0$
When $x = 0{:}\ y = 6(0) = 0$ (Inconclusive)
Use $\frac{dy}{dx}\ test{:}\ \frac{dy}{dx} = 3x^2$
$\Rightarrow 3(-0.1)^2 = 0.03\ /\ 3(0)^2 = 0\ /\ 3(0.1)^2 = 0.03$

x	-0.1	0	0.1
$\frac{dy}{dx} = 3x^2$	+	0	+
Shape	/	-	/

Therefore (0, 0) is a point of inflexion.■
Class 9 Limit: (3 types of discontinuity)
Removable Discontinuity **(Limit Exist)**
Jump Discontinuity **(Two Separate Limit)**
Infinite Discontinuity **(Limit Does Not Exist)**
Show whether the following function is continuous or not.
$f(x) = \begin{cases} \{x^2 & \text{for } x < 1\ (\to\to 1^-) \\ \{0 & \text{for } x = 1 \\ \{2 - (x-1)^2 & \text{for } x > 1\ (\leftarrow\leftarrow 1^+) \end{cases}$
When $x = 1,\ f(1) = 0$
When $x < 1,\ \lim_{x\to1^-} f(x) = \lim_{x\to1^-}(x^2) = (1)^2 = 1$
When $x > 1,\ \lim_{x\to1^+} f(x) = 2 - ((1) - 1)^2 = 2$
The $\lim_{x\to1^-} f(x)$ exists and $\lim_{x\to1^+} f(x)$ exists, however $\lim_{x\to1^-} f(x) \neq \lim_{x\to1^+} f(x)$ as $1 \neq 2$. Hence, $\lim_{x\to1} f(x)$ does not exist.
Therefore, the function is discontinuous at $x = 1$ as $\lim_{x\to1} f(x)$ does not exist. It is a Jump Discontinuity as it jumps from one value to another at that point.■
$\boldsymbol{f(x) = \frac{|x|}{x}}$ **When $x < 0$, $\lim_{x\to0^-}\frac{-2x}{x}$ / When $x < 0$, $\lim_{x\to0^+}\frac{2x}{x}$**
Example for Removable Discontinuity Conclusion
Since $\lim_{x\to1^-} f(x)$ exists and $\lim_{x\to1^+} f(x)$ also exists as $\lim_{x\to1^-} f(x) = \lim_{x\to1^+} f(x) = 2$ then $\lim_{x\to1} f(x) = 2$ exists. However, when $x = 1, f(1) = 5$. Therefore, the function is a removable discontinuity as f(1) must be redefined to $f(1) = 2$ for it to be continuous at $x = 1$.
L^1 Hospital Rule: $\frac{dy}{dx}$ Limit if: $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{0}{0}\ or \frac{\infty}{\infty}$
$\lim_{x\to0}\frac{e^x-1}{x^2+\sin x} \overset{L^1H}{=} \lim_{x\to0}\frac{e^x}{2x+\cos x} = 1$
Squeezing Theorem: $h(x) \leq f(x) \leq g(x)$ if h(x) = g(x) = L
$Therefore\ by\ Squeezing\ Theorem \Rightarrow \lim_{x\to a} f(x) = L$
Sin and Cos are bounded, $-1 \leq sin(x)\ /\ cos(x) \leq 1$

Class 10/11 (Matrix):	Homogeneous Coordinates
Scaling: $\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}$	Scale: $\begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & 1 \end{pmatrix}$
No Scale/Shear: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	Rotate: $\begin{pmatrix} \cos Q & -\sin Q & 0 \\ \sin Q & \cos Q & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Rotate About Origin: $\begin{pmatrix} \cos Q & -\sin Q \\ \sin Q & \cos Q \end{pmatrix}$	$\begin{pmatrix} \cos Q & \sin Q & 0 \\ -\sin Q & \cos Q & 0 \\ 0 & 0 & 1 \end{pmatrix}$
If "Clockwise" stated: $\begin{pmatrix} \cos Q & \sin Q \\ -\sin Q & \cos Q \end{pmatrix}$	Reflection: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Reflect about: x-axis (Flip vertically): $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
y-axis (Flip vertically): $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	Swapping: $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Standard Swap $y = x$: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Swap and Invert about origin $y = -x$: $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	Shearing: $\begin{pmatrix} x & 1 & 0 \\ 1 & y & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Shearing (x – Move Top, Y – Move Right Wall): $\begin{pmatrix} 1 & x \\ y & 1 \end{pmatrix}$	

Scale followed by Rotate followed by Translate: $\boldsymbol{T \circ R \circ S}$
$\boldsymbol{T = \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \to (T \circ R \circ S)\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = New\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = (x, y)}$
Determinant (D) Square Matrix: 2 by 2 = ad-bc $\boldsymbol{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}$
Minor of the Entry: (M_{ij}) means you ignore the Row and Column numbers of the (i j) in the current (M_{ij})
Cofactor:$(C_{ij}) = (-1)^{i+j}(M_{ij})\ (i + j = Even\ it\ is\ Positive)$
$\boldsymbol{Co(A) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}}$ **D=** $\boldsymbol{|A| = (aij)(Cij) + (aji^n)(Cij^n)}$
Adjoint: Inverse (i) Row (j) Column / $\boldsymbol{A^{-1} = \frac{1}{D}(adj(A))}$
Gaussian Elimination (GE) (R=Row) nR_n / $R_n \leftrightarrow R_n$/ $nR_n \pm R$

After that use Back Substitution to find x, y, z
Identity Matrix
$A \cdot A^{-1} = I$
$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Cramer's Rule
$\boldsymbol{A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{b} = \begin{pmatrix} n \\ m \end{pmatrix}}$
Find $x = \frac{\begin{pmatrix} n & b \\ m & d \end{pmatrix}}{|A|}$
Find y= $\frac{\begin{pmatrix} a & n \\ c & m \end{pmatrix}}{|A|}$
Vector Length:
$|\underbrace{a}| = \sqrt{a^2 + b^2}$
Gradient $= \frac{y}{x}$
Unit Vector:
$\hat{V} = \left(\frac{x}{|\underbrace{a}|}, \frac{y}{|\underbrace{a}|}\right)$
Free Vector for new position (AB) = New Position (x, y) – Old position (x, y)
$\boldsymbol{\overrightarrow{AB} = \overrightarrow{NB} + \overrightarrow{OA}}$
Parallel Vectors: Means same Gradient or both Vectors are Equal $\boldsymbol{\underbrace{a} = k\underbrace{b}}$
Class 12:
Ungrouped Data: (Raw list of numbers)
Mean: $(\bar{x}) = \frac{xi}{n}$
Median: Mid $= \frac{xm+(xm+1)}{2}$
Mode: is most frequent data
Range: Highest – Lowest
Population Variance: $\boldsymbol{\sigma^2} = \frac{(xi-(\bar{x}))^2}{n}$
Sample Variance: $\boldsymbol{S^2} = \frac{(xi-(\bar{x}))^2}{n-1}$
Standard Deviation: $\boldsymbol{\sqrt{\sigma^2}\ or\ \sqrt{S^2}}$
Grouped Data: (Table)
(fi) frequency of each group
Take the midpoint of each group score (mi) for the **Mean**: $(\bar{x}) = \frac{fimi}{n}$
Total people (N)=40
Median class people (M)=14
Number of people before Median Class (B)=16
Width of median Class Score (W)=10
Median class score: 30 to 39
(W)=29.5 to 39.5
Lowest boundary score (L): 29.5
Median: $L + \left(\frac{\frac{N}{2}-B}{M}\right) \times W$
$= 29.5 + \left(\frac{20-16}{14}\right) \times 10 = 32.36$
Group before Median class=(P)
Group after Median class=(A)
Mode: $L + \left(\frac{M-P}{2M-P-A}\right) \times W$
Variance: Take the midpoint of each group score (mi): $\boldsymbol{\sigma^2} = \frac{fi(mi)^2}{n} - (\bar{x})^2$
Probability: $\frac{n!}{c!\,p!\,q!}$
$0 \leq (P) \leq 1$
Total Outcome of a 6 and 4 sided Dice is **6 × 4**